Pore classes for the micromechanics calcuations

Input for MMM runs

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13 October 2026

Update for T. Barani

Jernkvist 2019, OperaHPC 5.2: rupture criterion

$$\hat{s}\left(p - \frac{2\gamma}{R} - P_h\right) \geq \sigma_R + P_h \quad \Longleftrightarrow \quad p \geq P_{cr} = P_h + \frac{2\gamma}{R} + \frac{\sigma_R + P_h}{\hat{s}}$$

Jernkvist 2019 (Olander)

$$P_{cr} = \frac{2\gamma}{R} + \frac{\sigma_{hbs}^{cr}(1-\xi) + P_h}{\xi},$$

$$\sigma_{hbs}^{cr} = 21 \text{ MPa} \quad \hat{s} = \frac{\xi}{1-\xi}$$

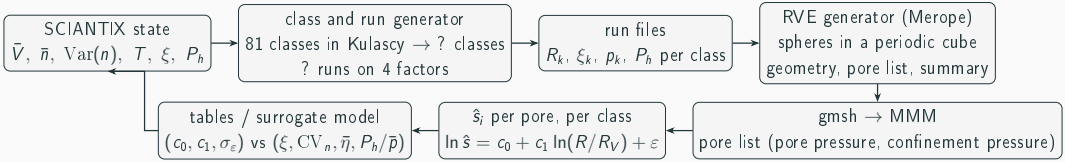
OperaHPC D5.2

$$\sigma_l^{max}\big|_i \geq \sigma_R = 275 \text{ MPa},$$

$$\hat{s}_i = \frac{\sigma_l^{max}|_i(1, p_{hyd}) + p_{hyd}}{1 - p_{hyd}}$$

Scope and chain of tools

Goal: replace the mean-field stress factor $\xi/(1-\xi)$ of Jernvist by tables computed with MMM on RVEs that reproduce the pore classes that will be seen by SCIANTIX.



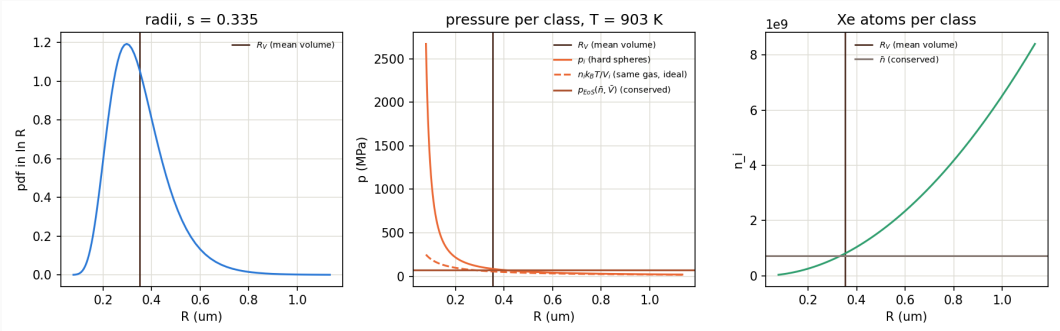
Pore population generator I

Porosity model from SCIANTIX: $\bar{V}$ (R_V), $\bar{n}$ = Xe-atoms/pore, $\mathrm{Var}(n)$, N_p, $\xi = N_p \bar{V}$.

log-normal R (K15) : 81 nodes $x_i \in [-4, 4]$, $w_i \propto e^{-x_i^2/2}$, $R_i = R_V\, e^{s x_i} \Big/ \left(\sum_j w_j e^{3 s x_j}\right)^{1/3}$ $\left(\sum_i w_i V_i = \bar{V}\right)$

Xe/pore distribution : $n_i = \bar{n}\, R_i^a \Big/ \sum_j w_j R_j^a$, $a = 2$ (gas density $\propto 1/R$), s from $\sum_i w_i (n_i - \bar{n})^2 = \mathrm{Var}(n)$

pressure distribution : $p_i = \dfrac{n_i k_B T}{V_i} Z(\eta_i)$, $\eta_i = \min\left(\dfrac{n_i v_{Xe}(T)}{V_i}, 0.65\right)$ hard spheres



NFIR, end of base irradiation: $s = 0.33$ (measured about 0.3, K15 0.356), radii 0.08–1.1 μm, p_i from 2.7 GPa to 30 MPa.

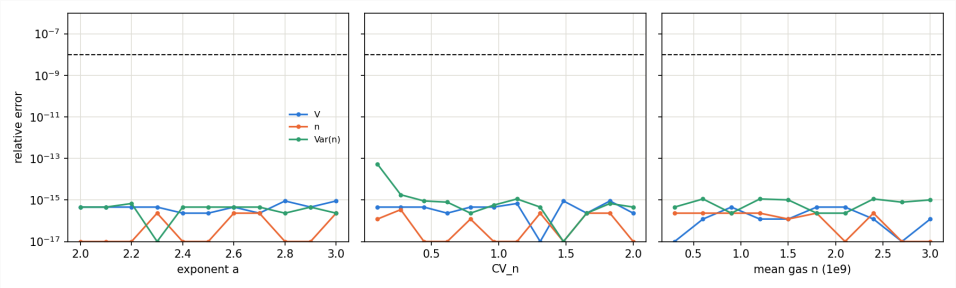
Pore population generator II

$$\sum_i w_i V_i = \bar{V} \qquad \text{exact by construction}$$

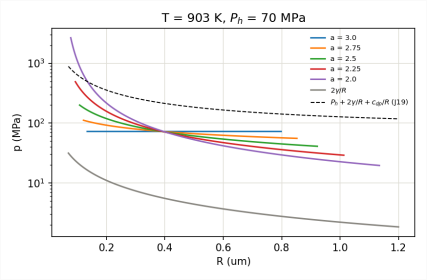
$$\sum_i w_i n_i = \bar{n} \qquad \text{exact by construction}$$

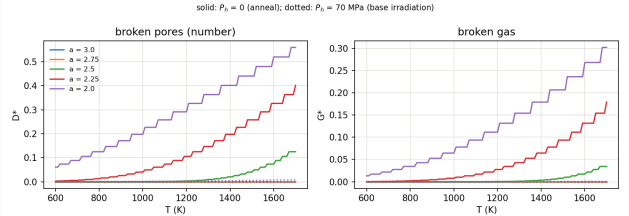
$$\mathrm{CV}_n^2(s, a) = \frac{\sum_i w_i e^{2asx_i}}{\left(\sum_i w_i e^{asx_i}\right)^2} - 1 = \frac{\mathrm{Var}(n)}{\bar{n}^2} \qquad \text{root in } s$$

Mean pressure (pressure of the volume-weighted density): $p_{EoS}(\bar{n}, \bar{V}, T)$ is conserved.



Pore population generator III: Xe atoms/pore





The moments hold for every a; what changes is the width s, the pressure spread and which classes break: small pores first for $a < 3$, while for $a = 3$ uniform pressure.

Reduction of the parameter space

MMM calculations will be scale-free and linear in the load: R_V and $\bar{p}$ only set the scale. With $p^* = p\, v_{Xe}/k_B T$ the EoS has no explicit T,

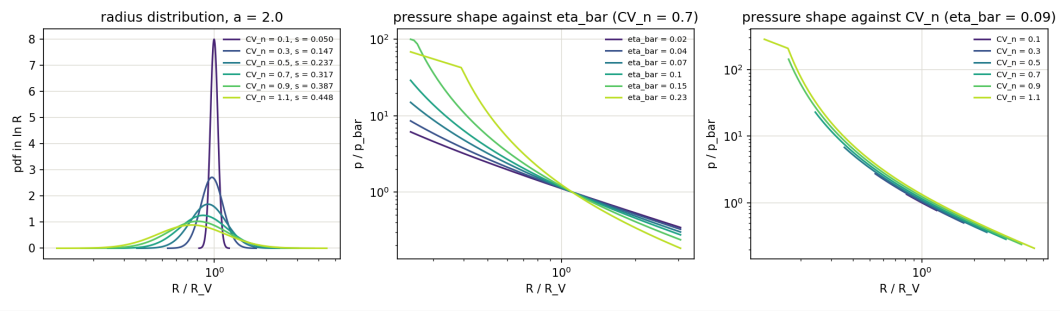
$$p_i^* = \eta_i Z(\eta_i), \qquad \bar{\eta} = \frac{\bar{n}\, v_{Xe}(T)}{\bar{V}}, \qquad \mathrm{CV}_n = \frac{\sqrt{\mathrm{Var}(n)}}{\bar{n}}$$

so in $x_i = R_i/R_V$ the shape $p_i/\bar{p}$ and the weights w_i depend on $(\bar{\eta}, \mathrm{CV}_n, a)$ only.

factor	sets
ξ	porosity, pore distances
s (from CV_n)	R_k/R_V and the shares ξ_k/ξ
$\bar{\eta}$	with s, the pressure profile $p_k/\bar{p}$
$P_h/\bar{p}$	net load $p_k - P_h$ of every class

How to cover the parameter space for Merope + MMM run? Ranges and levels to be defined, LHS could be a way to define the different combinations.

Parameter influence



Merope

For every run file:

1. read the 4 columns, sort the classes by decreasing radius, check that P_h is one value
2. expected pores per class $= \xi_k / V_k \cdot L^3$
3. throw the spheres by RSA in a periodic cube, one (radius, porosity) pair per class, with a repulsion distance
4. each sphere gets the class with the nearest radius, hence its pressure; one phase id per sphere, not meshed
5. write, in one folder per run, seed and repulsion: the pore list (id, x, y, z, R, p), a summary (obtained vs asked porosity per class, pore counts), the geometry for gmsh

Fixed in the generator:

cube	30 μm	$\approx 86\, R_V$
repulsion	0.3 μm	to verify a posteriori
mesh	0.15 μm	$> R_{min}/2$, to be checked
seed	5	

Status

- Xe atoms / pore - distribution is fixed by a.
- Dislocation punching pressure.
- LHS sampling and parameter range.
- Pore count (smallest class should be represented).
- Smallest classes should be resolved,by the mesh size.
- Repulsion distance should scale with the radius?
- 1 seed per run.
- Merope + MMM calculations on few runs to evaluate computational cost.
- Surrogate model: training (LHS sampling) and validation dataset (additional cases).

Other stuff

- **Form for Marjorie**: filled in; to confirm the affiliation to put (PLEIADES or UT) and the phone number field.
- **Abstract**: review requested before submission.
- **Lectures**: indicative dates of absence 3–5 November.
- **Transparant online meeting at the end of the month**: I would ask for a day off this week, working from home for the meeting.